

mm-Wave Phase Interpolator via Frequency Doubling

70 GHz Clock Generation

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Outline

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Motivation

Why mm-Wave Phase Interpolation?

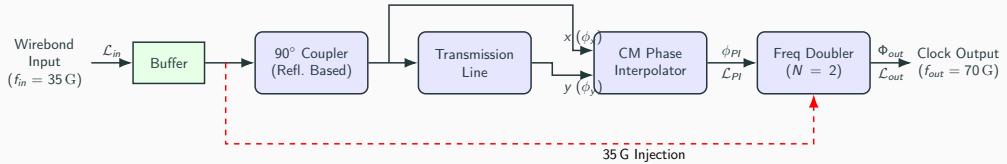
- mm-wave bands (24 GHz to 100 GHz) are key for 5G NR, automotive radar, and high-speed backplane links
- Fine, monotonic phase control is required for beamforming and CDR
- Conventional PI topologies face bandwidth and linearity limits above 20 GHz

Key insight

Operate the interpolating core at $f_0/2$, then double — relaxes speed while maintaining fine phase resolution at mm-wave.

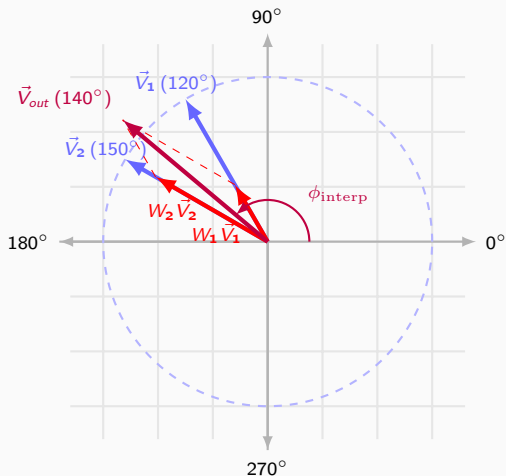
System Architecture

Top-Level Architecture



Analysis

Phase Interpolation — Phasor View



Interpolation steps:

1. **Coarse:** select boundary phasors $\vec{V}_1$, $\vec{V}_2$ (here 120°, 150°)
2. **Fine:** adjust tail-current weights W_1 , W_2
3. **Sum:** resultant angle follows

$$\frac{W_2}{W_1} = \frac{\sin(\phi - \phi_1)}{\sin(\phi_2 - \phi)}$$

Key constraint

Boundary phases must span the desired range; weight nonlinearity sets INL.

Phase Noise Through the Doubler

Multiplication rule ($\times N$):

$$\mathcal{L}_{\text{out}} = \mathcal{L}_{\text{in}} + 20 \log_{10} N + \mathcal{L}_{\text{add}}$$

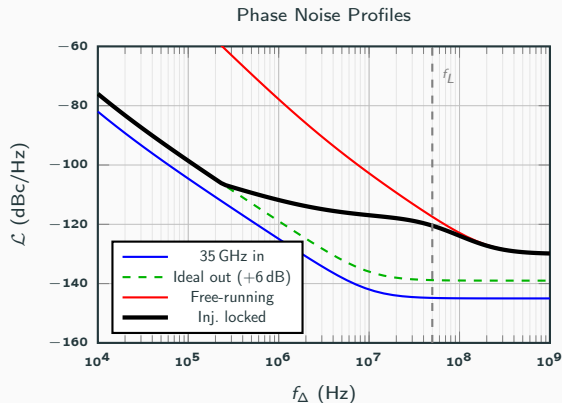
For $N = 2$: irreducible +6 dB degradation.

Jitter preservation:

$$\sigma_{t,\text{out}} = \frac{N\sigma_{\phi}}{2\pi Nf_{\text{in}}} = \sigma_{t,\text{in}}$$

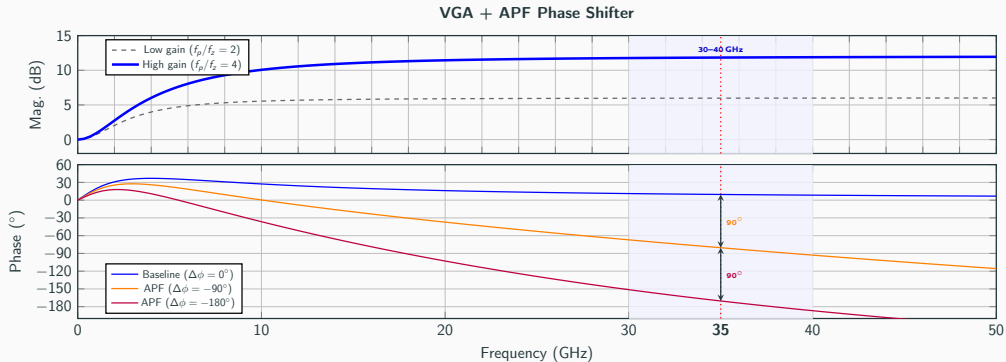
Absolute jitter is *ideally conserved*.

Injection locking clamps output noise to $\mathcal{L}_{\text{in}} + 6 \text{ dB}$ within f_L .



Circuit Implementation

VGA & Phase Shifter — Bode Plot



[CML schematic — placeholder]

Results

[Post-layout simulation results — placeholder]

Conclusion

- Frequency-doubling PI architecture relaxes bandwidth on the interpolating core at mm-wave frequencies
- Ideal frequency doubling preserves absolute jitter (phase noise degrades by exactly +6 dB)
- Injection locking confines output noise to the ideal bound within the locking bandwidth f_L
- APF-based phase shifter provides 180° continuous tuning across the 30 GHz to 40 GHz band

Questions?